Development and robustness: slower trading helped, but did not establish a durable edge

Report 1 of 3 | Development | 1 January 2022-31 December 2023

Per-coin P&L and joint basket alpha/betas, with confidence intervals, for all nine frozen strategies.

Executive Summary

The main finding is failure to generalize across periods

Follow-up orientation through 30 September 2026, not an input to the original development selection. All four development-selected Track A strategies lost in 2024-2025; their raw Sharpes ranged from **-0.64 to -0.22 (365-day)**, or **-0.53 to -0.18 (252-day)**. All five validation-informed Track B strategies lost compounded net return in January-September 2026. B06 still had slightly positive arithmetic raw Sharpe, **0.05 / 0.04 (365 / 252)**; a compounded loss does not require negative Sharpe.

This pattern is **consistent with overfitting or weak transfer, not proof of one cause**: baskets, settings and cadence differ, and Track B also adds long/cash trend. A03's **+31.38%** in 2026 and raw Sharpe **2.32 / 1.93** do not erase its **-22.31%** validation loss. No new winner or reliable all-period strategy is established.

Sharpe convention: tables show **365 / 252** on the **same UTC daily series including weekends**. The alternative is S252 = S365 × √(252/365) = 0.8309097 × S365; it does not drop weekends. An ≈ marks a 252-day value available only from a rounded 365-day source. The primary 365-day results, inference, rankings, CAGR and alpha are unchanged. Full-precision convention comparison.

Separate fixed all-22 studies did not establish persistence. Development / validation / Jan-Sep 2026 net returns were **+5.61% / -17.43% / -20.94%** for momentum and **-33.59% / -15.66% / +9.91%** for reversal. Neither produced positive net returns in all three periods. These remain separate fixed studies, not a new direction selection or fresh independent confirmation. Study results, separate tests and chronology disclosures.

Development evidence

- Only **106,302 of 1,117,935 paths (9.51%)** had positive compounded net return at 20 bps. The full-grid median was **-71.03%**, versus **-7.89% before modeled fees**. Fast turnover was a major obstacle, not the only one.
- Slower trend definitions had better development distributions: median net returns were **+4.05% long/cash** and **+22.04% theoretical long/short**. These are unequal family grids, not evidence that trend universally dominates cross-sectional strategies.
- The apparent edge was uneven. **Every long/cash path lost in 2022 and gained in 2023**; only **39.15% of long/short paths** were positive in both years. Basket and event concentration mattered.
- This stage supported **development robustness checks before selection**, not a discovery claim. All unsuccessful configurations remain in the results; the 169-definition space is research-defined, not an independent discovery test.

Study scope

The study evaluates the complete configuration space across systematically enumerated liquidity-tier baskets. The configuration space and historical universe are research-defined; this large correlated search does not provide independent confirmation of an edge.

The complete study covered **6,615 eight-coin baskets × 169 definitions**, or **1,117,935 primary basket-configuration pairs**. Seven fee/delay scenarios produced **7,825,545 result rows**. All were evaluated; **1,000 genuine all-cash paths had undefined Sharpe**, not zero Sharpe. Every basket had at least one positive development result, which is unsurprising after a large correlated search.

Development was 2022-2023. Validation was reserved as 2024-2025 and 2026 as a later evaluation; neither was used in the original development grid or development-only selection. The cross-period opening is follow-up orientation, not a selection input. Reports 2 and 3 cover those subsequent stages.

Separate fixed all-22 studies did not establish persistent performance

These are **two separately approved simplicity studies**, responding to the proposal to remove basket/parameter selection. Each fixes one 28-day cross-sectional rank-demean rule on all 22 eligible economic slots, with Monday 00:00 UTC rebalancing and no buffer. Momentum buys relative winners and shorts losers; the independent reversal study buys relative losers and shorts winners. No direction is selected, no tests are pooled, and neither changes the nine frozen strategies.

Momentum gained modestly in development, then lost in validation and partial-year 2026. **Reversal** lost in development and validation; its positive partial 2026 does not establish persistence. Neither produced positive net returns in all three periods. The pattern across periods—not the best isolated result—is the finding.

Standalone momentum: unchanged 20 bps / lag1 primary

Period	Net return	Net Sharpe 365 / 252	Hourly max DD	Annual full-L1 turnover
2022–2023 development	5.61%	0.23 / 0.20	29.53%	33.58×
2024–2025 validation	-17.43%	-0.38 / -0.31	26.56%	34.91×
Jan–Sep 2026 (273 days)	-20.94%	-1.70 / -1.41	22.06%	39.29×

Standalone reversal: unchanged 20 bps / lag1 primary

Period	Net return	Net Sharpe 365 / 252	Hourly max DD	Annual full-L1 turnover
2022–2023 development	-33.59%	-0.87 / -0.72	43.69%	33.58×
2024–2025 validation	-15.66%	-0.32 / -0.27	31.57%	34.91×
Jan–Sep 2026 (273 days)	9.91%	0.81 / 0.67	11.10%	39.29×

2022–2023 development: additional primary risk and growth

Study	Gross return	CAGR estimate	Net volatility √365	Daily max DD	Hourly DD episode
Momentum	20.79%	2.77%	21.02%	28.19%	Peak 2022-05-11 Trough 2022-09-12 Recovered 2023-12-21 588.3 days underwater
Reversal	-24.02%	-18.51%	21.03%	43.26%	Peak 2022-09-12 Trough 2023-12-26 Not recovered; censored 475.5 days underwater

Measurement. Net returns include transaction fees; gross returns exclude them. Both Sharpe annualization conventions use the same UTC daily observations including weekends: $S252 = S365 \times \sqrt{(252/365)}$. CAGR, volatility and alpha use 365 days. Daily drawdown and original hourly drawdown use different endpoint clocks; both include initial NAV. Turnover is full-L1 traded-weight multiples including entry, forced events and liquidation. Each two-year return is compounded; CAGR is annualized.

2022–2023 development: uncertainty remains material

The saved **net daily-mean HAC27 t-statistics are descriptive.** Net Sharpe intervals below are **marginal and conditional** on the fixed rule and historical sample, not simultaneous or selection-adjusted evidence for all prior research. No new hypothesis tests or strategy selections are performed.

2022–2023 development: descriptive net-return uncertainty

Study	Net-mean HAC27 t	Net Sharpe 95% CI: 365 / 252
Momentum	0.341	365 [-1.12, 1.59] 252 [-0.93, 1.32]
Reversal	-1.260	365 [-2.30, 0.49] 252 [-1.91, 0.41]

2022–2023 development: descriptive BTC sensitivity, marginal HAC7 intervals

Study	Alpha/year [95% CI]	BTC beta [95% CI]
Momentum	5.14% [-23.91%, 34.18%]	-0.018 [-0.065, 0.028]
Reversal	-18.43% [-47.50%, 10.64%]	0.017 [-0.030, 0.064]

The cross-period result is weaker than an isolated profitable window

Neither standalone rule was profitable after costs in all three periods. Momentum's hourly drawdowns stayed below the fixed 40% guardrail; reversal's development hourly drawdown reached **43.69%**. These observations do not support a deployment-ready, persistent edge.

Uncertainty methods: Bartlett HAC27 covariance with n/(n–1) for the descriptive mean t-statistic; 10,000 circular 28-day bootstrap draws per period, PCG64 seed 20261020 reset per period. Net Sharpe intervals are percentile intervals from the saved draws; the 252-day endpoints are the same positive rescaling of the 365-day endpoints. All draws were valid. Fees/trades are not reinserted at resampled block boundaries.

BTC coefficients: alpha is 365 times the daily arithmetic intercept, not CAGR. Full-calendar HAC7 intervals are descriptive. These are BTC-only estimates, **not joint 22-coin regressions**, and are not selection gates.

Preserve the procedural distinction

Momentum: its code/settings seal preceded the first all-22 simulation, but existing verification tests had already decoded packaged panels and recalculated saved-nine derived statistics. Strict pre-panel-access compliance is **false**; the chronology exception and corrections remain in its evidence. **Reversal:** its own freeze preceded the first new-study panel decode; strict pre-access checks passed.

All three historical windows had already been examined in the wider project. These additional studies **are not fresh independent OOS confirmation**. A failed rule here does not prove all momentum or reversal strategies fail. No basket, lookback, cadence, direction or fee sensitivity is promoted after seeing these outcomes.

2022–2023 development: all predeclared sensitivities retained

Primary remains **20 bps / lag1**. Lag1 is idealized observed-close execution, not a guaranteed one-hour wait; lag2 adds one hour. Each independent study completed **21** scenario-period keys, with no missing or failed slots. These tables retain all seven scenarios for this report's period; no sensitivity becomes a new selection.

Standalone momentum: 2022–2023 development

Cost bps / lag	Net return	Net Sharpe 365 / 252	Hourly max DD
0 / 1	20.79%	0.56 / 0.46	27.78%
5 / 1	16.80%	0.48 / 0.39	28.22%
10 / 1	12.95%	0.40 / 0.33	28.66%
15 / 1	9.22%	0.31 / 0.26	29.10%
20 / 1	5.61%	0.23 / 0.20	29.53%
40 / 1	-7.68%	-0.08 / -0.07	32.41%
20 / 2	5.10%	0.22 / 0.19	29.95%

Standalone reversal: 2022–2023 development

Cost bps / lag	Net return	Net Sharpe 365 / 252	Hourly max DD
0 / 1	-24.02%	-0.55 / -0.46	38.88%
5 / 1	-26.54%	-0.63 / -0.52	40.12%
10 / 1	-28.96%	-0.71 / -0.59	41.33%
15 / 1	-31.31%	-0.79 / -0.65	42.52%
20 / 1	-33.59%	-0.87 / -0.72	43.69%
40 / 1	-41.95%	-1.19 / -0.99	48.26%
20 / 2	-33.28%	-0.85 / -0.71	43.93%

Full universe, data and supporting evidence

Fixed historical-liquidity slots: BTC, ETH, SHIB, BNB, SOL, XRP, DOT, SAND, legacy LUNA/LUNC, AVAX, DOGE, ADA, FTM/S, TRX, MATIC/POL, MANA, FIL, GALA, LINK, AXS, LTC and VET. Eligible members are reranked together; BTC/ETH are ordinary members, not fixed sleeves. No reconstitution. Inherited inventory/survivor limitations remain.

Source: Binance Data Vision spot-USDT hourly archives; **43,824** normalized endpoints from **1 October 2021 01:00 UTC through 1 October 2026 00:00 UTC**, including warm-up, with seven pinned panels and explicit lifecycle/halt masks. No data were downloaded, gaps filled or windows shortened. Full data manifest. Shorts are theoretical; borrow, funding, margin, impact, capacity, realistic fills and ordinary drift-maintenance are unmodelled. USDT/USD parity is assumed.

Complete three-period evidence: Momentum results, scenarios, uncertainty. Reversal results, scenarios, uncertainty.

The nine frozen strategies: development performance

2022–2023; primary 20 bps costs, all nine fixed IDs. This nine-case comparison is separate from the full-grid family distributions; it is not a new selection. Track A was selected using development. Track B identities were selected later using development plus validation, so these development values describe those later-fixed identities, not development-only candidates.

All nine fixed strategies: net performance

ID	Net return	CAGR estimate	Net Sharpe 365 / 252	Annual full-L1 turnover	Hourly max DD
A01	165.36%	62.90%	1.84 / 1.53	31.44×	23.25%
A02	252.94%	87.87%	2.03 / 1.69	49.58×	58.37%
A03	64.38%	28.21%	1.19 / 0.99	27.62×	19.79%
A04	314.22%	103.52%	1.43 / 1.19	50.50×	51.35%
B01	0.34%	0.17%	0.14 / 0.12	81.69×	37.34%
B03	1.05%	0.53%	0.15 / 0.12	27.75×	27.64%
B04	44.11%	20.04%	0.63 / 0.52	21.31×	61.98%
B05	12.60%	6.11%	0.35 / 0.29	11.88×	36.50%
B06	35.86%	16.56%	0.55 / 0.46	22.88×	48.23%

Net results include 20 bps transaction costs. Returns are full-period compounded changes; CAGR and Sharpe are annualized estimates. Both Sharpe conventions use the same UTC daily returns including weekends: S252 = S365 × $\sqrt{(252/365)}$. Hourly maximum drawdown and full-L1 turnover retain the original accounting clock. BTC alpha is an arithmetic regression intercept annualized with 365, not CAGR.

The basket rule used historical liquidity within an inherited pool

At the 1 January 2022 decision, coins were ranked by **median daily USDT quote volume across 90 observations**, with endpoints from **3 October through 31 December 2021** and a **24-hour information lag**. This used the inherited historical top-30 research universe, not an unrestricted census of every exchange listing. The basket campaign used ranks **1-22** only, with no reconstitution.

BTC and ETH were fixed. Each basket took two coins from ranks 3-9, two from 10-16 and two from 17-22: **C(7,2) × C(7,2) × C(6,2) = 6,615** baskets. Every signal was recomputed inside its own eight-coin basket; thirty-coin portfolio weights were not sliced.

Rank is historical liquidity rank; volume is millions of USDT per day. Quote volume measures activity, not a demonstrated executable spread or trading capacity.

Fixed slots

Rank	Pair	Median volume (USDT m)
1	BTCUSDT	2384.13
2	ETHUSDT	1724.40

Tier 1: choose two from ranks 3-9

Rank	Pair	Median volume (USDT m)
3	SHIBUSDT	718.95
4	BNBUSDT	605.06
5	SOLUSDT	463.10
6	XRPUSDT	385.78
7	DOTUSDT	353.16
8	SANDUSDT	303.11
9	LUNAUSDT	285.00

Tier 2: choose two from ranks 10-16

Rank	Pair	Median volume (USDT m)
10	AVAXUSDT	284.44
11	DOGEUSDT	272.77
12	ADAUSDT	261.92
13	FTMUSDT	244.35
14	TRXUSDT	228.72
15	MATICUSDT	223.37
16	MANAUSDT	208.63

Tier 3: choose two from ranks 17-22

Rank	Pair	Median volume (USDT m)
17	FILUSDT	203.47
18	GALAUSDT	183.98
19	LINKUSDT	140.30
20	AXSUSDT	121.01
21	LTCUSDT	116.42
22	VETUSDT	103.97

LUNA denotes the **legacy Terra/LUNC economic slot, not Terra 2.0**. A suspension or unavailable slot remained cash under the preserved lifecycle policy; it did not trigger replacement by a current survivor.

Six families, with different economic exposures

Momentum and reversal: 78 definitions each. The lookbacks were 1, 2, 4, 8, 12, 24, 48, 72, 168, 336, 672, 1,440 and 2,160 hours; rebalancing was every 1, 4, 12, 24, 72 or 168 hours. Cross-sectional trailing-return ranks were demeaned and scaled to full-L1 gross exposure one, with dollar neutrality. Reversal used the opposite sign. Dollar neutrality does not ensure BTC-beta neutrality.

Buffered momentum: four definitions. The 14/28-day lookbacks crossed daily/weekly decisions. The entire rebalance was skipped if proposed full-L1 target change was below 0.25; otherwise the full target was executed. Four matching unbuffered momentum definitions were already in the 78-case grid.

Time-series trend: four long/cash and four theoretical long/short definitions. The same 14/28-day lookbacks crossed daily/weekly decisions. Each coin had an unscaled 1/8 slot: positive signal meant +1/8 for long/cash, otherwise cash; long/short used sign(trailing return)/8. Remaining active positions were not rescaled.

One fixed BTC-factor residual reversal. For seven non-BTC coins, monthly formation used 90 daily observations ending 24 hours before refit: earliest 30 to estimate BTC return beta, subsequent 60 endpoints to calibrate log-price spread mean/standard deviation. Eligibility required beta in [0.1,5], positive variance, Holm-adjusted ADF significance at 5%, and AR(1) half-life 1-20 days. Coefficients stayed frozen during the month.

Residual entries required $2 \le |z| < 4$, toward the mean. Exits used the ±0.5 mean boundary, adverse ±4 stop, or 20 calendar days held; no same-day reentry. Month-start forced cash/refit, entries from the next day. Each spread had 1/7 of capital including its BTC hedge; hedges were netted before costs. Ineligible spreads stayed cash. This model was not necessarily dollar- or beta-neutral.

Common execution and accounting

Hourly Binance spot-USDT history began with warm-up on 1 October 2021. Each evaluated period started in cash and paid entry and terminal liquidation charges. Elapsed-time schedules and the inherited Monday epoch were preserved; weekly is not every seventh available row.

Primary costs were **20 bps one way × full-L1 target changes**, with 0/5/10/15/40 bps sensitivities. **Lag1 means an idealized observed-close fill, not one hour of elapsed execution latency**; lag2 adds one hour. Halt bridges, missingness, forced exits and partial event accounting were preserved. There was no silent filling or sample shortening.

Trend had better development medians, but family grids were not comparable experiments

At 20 bps, unbuffered momentum and reversal had strongly negative median paths. Trend and buffered momentum used only four slow definitions each, compared with 78 definitions in each unbuffered family. Thus the broad family comparison mixes signal form, cadence, lookback and exposure. It cannot establish a universal ranking of trading styles.

2022-2023 development: distribution medians across each family's complete grid

Family	Positive	Gross return	Net return	Sharpe (365 / 252)	Max DD
Momentum	9.51%	-1.44%	-60.86%	-1.53 / -1.27	72.36%
Reversal	2.62%	-21.03%	-83.81%	-2.31 / -1.92	85.73%
Buffered momentum	40.51%	24.51%	-6.99%	0.02 / 0.02	39.80%
Trend long/cash	56.30%	17.53%	4.05%	0.25 / 0.20	45.84%
Trend long/short	65.59%	58.16%	22.04%	0.46 / 0.38	58.26%
BTC residual reversal	11.34%	-0.46%	-0.58%	-0.64 / -0.53	0.81%

All return numbers are compounded **two-year returns**, not annual returns. Max DD is nonnegative loss depth. Turnover is annualized **full-L1 target-change multiples**, including entry/liquidation, not conventional half-L1 turnover or actual drift-adjusted traded notional. Column medians may correspond to different individual paths.

Family median annual full-L1 turnover was **268.69× momentum/reversal**, **62.26× buffered momentum**, **20.75× long/cash**, **41.56× long/short**, and **0.29× residual reversal**.

The residual model's small median loss and low turnover reflect sparse eligibility/trading, not a low-risk established edge. Its all-cash cases retain undefined Sharpe. The theoretical short books omit borrow/funding/margin constraints. Long/cash carries variable market participation and cash, so equal gross/net returns do not imply equal risk.

Read risk alongside return

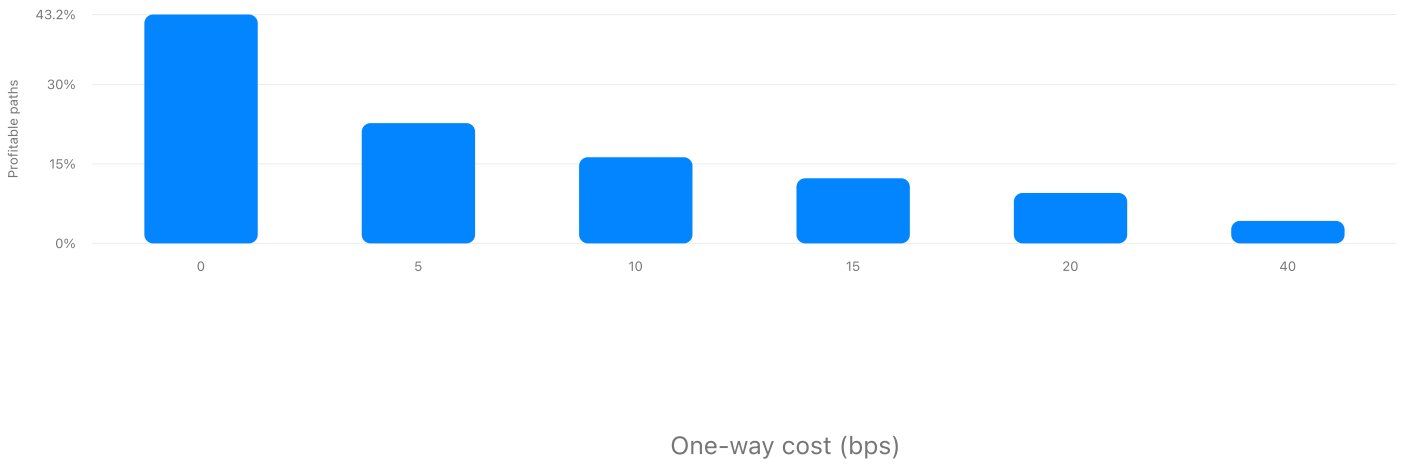
Across the entire unequal grid, the median CAGR was **-46.18%**, annual volatility **27.15%**, Max DD **76.85%**, and longest underwater span **708.6 days**. The median net Sharpe was **-1.77 (365-day) / -1.47 (252-day)** and annualized full-L1 turnover **189.23×**. These summarize individual research paths, not an implementable aggregate portfolio.

The results also retain Sortino, Calmar, expected shortfall, worst day/week, ulcer index, win rate, gain/loss ratio, time underwater, exposure, and BTC alpha/beta. No single maximum or aggregate score substitutes for these risks.

Fees destroyed much of the broad grid's already weak economics

At zero modeled fees, **43.17%** of development paths were profitable. At 20 bps that fell to **9.51%**, and at 40 bps to **4.24%**. The zero-cost median was already negative, so the problem was not solely execution costs. Gross return minus a simple percentage fee is not the net compounded result.

Full-grid profitable breadth under sampled fees



All retained fee/delay scenarios

Scenario	Profitable	Median return	Median Sharpe (365 / 252)
cost00_lag1	43.17%	-7.89%	-0.01 / -0.01
cost05_lag1	22.68%	-36.87%	-0.70 / -0.58
cost10_lag1	16.25%	-51.34%	-1.09 / -0.91
cost15_lag1	12.28%	-62.42%	-1.44 / -1.19
cost20_lag1	9.51%	-71.03%	-1.77 / -1.47
cost40_lag1	4.24%	-87.46%	-3.03 / -2.52
cost20_lag2	9.60%	-70.80%	-1.75 / -1.45

Slow schedules helped, but weekly was not automatically safer

The pre-existing four-clock intersection compared five nonfactor families at 14/28-day lookbacks and daily/weekly decisions on all 6,615 baskets. This removed unequal lookback/cadence coverage, but not exposure or allocation differences.

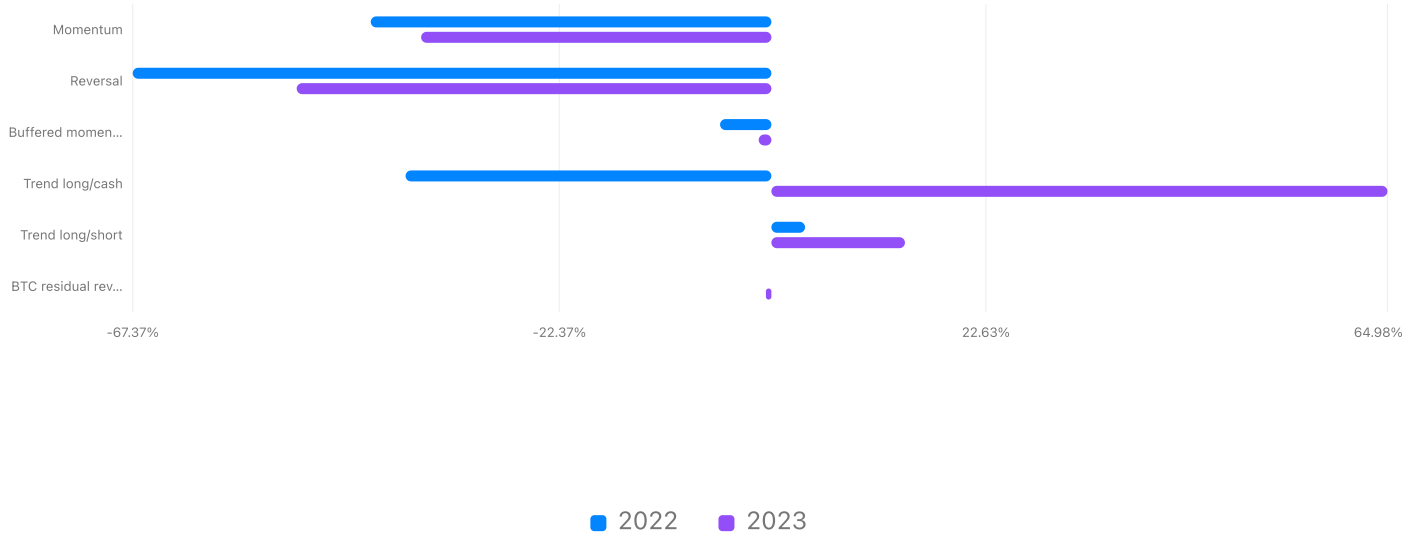
For 14-day long/cash, mean development return was **+1.70% daily versus -25.72% weekly**; for 28-day long/cash it was **+17.33% versus +26.81%**. These are means across matched basket sets, not portfolio returns. Thus “slow reduces costs” was supported, while “weekly always improves outcomes” was not.

The whole-target buffer improved paired median net return by **1.09 pp (14-day daily)** and **3.05 pp (28-day daily)**, but had approximately zero median effect for the two weekly clocks. It was a specific turnover intervention, not a universal signal improvement. The extra-hour scenario retained the same overall weak median economics.

A two-year profit did not imply success in both years

All 26,460 long/cash basket-definition pairs lost in 2022 and gained in 2023. Only 10,358 of 26,460 (39.15%) long/short pairs were positive in both. Across every family, just 44,982 of 1,117,935 paths (4.02%) were positive in both years. These highly correlated counts are descriptive, not independent replications.

Annual median net returns by family, 20 bps



Same-pair annual sign transitions

Family	Both positive	Only 2022	Only 2023	Neither
Momentum	4.78%	7.03%	9.26%	78.94%
Reversal	0.69%	5.35%	3.33%	90.63%
Buffered momentum	24.22%	14.07%	22.82%	38.89%
Trend long/cash	0.00%	0.00%	100.00%	0.00%
Trend long/short	39.15%	15.49%	22.06%	23.31%
BTC residual reversal	0.00%	0.00%	11.34%	88.66%

Annual figures were **slices of the same continuous development path**, not separate annual re-runs. No artificial January cash reset, entry or liquidation fee was inserted; natural scheduled trades/refits and their actual fees remained at their original endpoints.

Within-basket configuration Sharpe rank correlation between the two years had median **0.867** across 6,615 baskets. That high pooled rank correlation can reflect stable separation of very expensive versus slow definitions; it does not demonstrate persistence among profitable leaders or identify a robust parameter winner.

Nearby baskets were often similar; nearby parameters could still change the result

Parameter neighbors changed one adjacent declared lookback or rebalancing level, preserving the other coordinate, family, phase and buffer. Basket neighbors replaced exactly one nonfixed coin within its liquidity tier: **28 neighbors per basket**, sharing seven of eight coins. These were not independent tests.

The table conditions its neighbor-positive share on an already-positive center with finite neighbors; it does not eliminate failures from the full grid. Absolute return changes are medians across all centers and do not cancel opposite signed changes.

Declared local robustness, without selecting a plateau

Family	Param. positive	Basket positive	Param. |ΔR|	Basket |ΔR|
Momentum	50.03%	67.43%	14.67 pp	6.87 pp
Reversal	29.79%	54.56%	8.16 pp	4.63 pp
Buffered momentum	66.08%	73.15%	18.56 pp	14.85 pp
Trend long/cash	55.32%	89.71%	26.48 pp	8.12 pp
Trend long/short	62.37%	92.27%	61.17 pp	19.78 pp
BTC residual reversal	undefined	64.29%	undefined	0.21 pp

For positive momentum centers, the **mean fraction** of profitable parameter neighbors was **50.03%**; for reversal it was **29.79%**. The corresponding basket-neighbor fractions were **67.43%** and **54.56%**. Trend's basket fractions were higher, but the theoretical long/short median absolute parameter change was **61.17 percentage points** of two-year return. A profitable neighborhood was not an invariant signal.

The factor model has no parameter neighbor because it has only one declared definition; that comparison is **unavailable, not zero**. Near-zero signed perturbation summaries were not treated as proof of robustness. No neighbor threshold or candidate was selected in this development report.

Legacy LUNA exposure was a material event and basket dimension

There were **1,890 baskets with legacy LUNA/LUNC** and **4,725 without**. Long/short trend's median development return was **+52.33% with LUNA versus +14.36% without**. Reversal was worse with LUNA: **-92.52% versus -73.20%**. Other basket memberships also changed, so these are membership-conditioned associations, not causal coin-removal effects.

All-family basket composition comparison, 20 bps

Family	With LUNA return	Without LUNA return	With LUNA Sharpe (365 / 252)	Without LUNA Sharpe (365 / 252)
Momentum	-59.00%	-61.51%	-0.91 / -0.76	-1.80 / -1.49
Reversal	-92.52%	-73.20%	-2.14 / -1.78	-2.46 / -2.04
Buffered momentum	-3.88%	-8.30%	0.12 / 0.10	-0.04 / -0.03
Trend long/cash	4.73%	3.74%	0.25 / 0.21	0.24 / 0.20
Trend long/short	52.33%	14.36%	0.65 / 0.54	0.41 / 0.34
BTC residual reversal	-1.05%	-0.48%	-0.55 / -0.46	-0.78 / -0.65

Legacy LUNA net dollar contribution in May 6-20, 2022



The event-window mean LUNA contribution was **+$0.1448** for long/short trend and **-$0.1207** for reversal, per original $1 continuous NAV and averaged over **all family paths**, including structural zeros when LUNA was absent. This is reconciled traded-coin P&L accounting, not a counterfactual return after deleting the event. The event window overlaps 2022 and the full path and must not be added twice.

The forced exit used the last **00:40 UTC partial quote on 13 May**, in the 01:00 accounting bucket: 40 minutes of LUNA exposure and possible temporary orphan hedge. This idealized liquidation does not establish executable short profit, borrow availability, or treatment of Terra 2.0/airdrop liabilities.

What these development results did and did not establish

Data source and complete study dataset. Binance public Data Vision (**data.binance.vision**) spot-USDT **1-hour klines** supply prices and quote volumes. The normalized UTC grid has **43,824 hourly close endpoints**, from **1 October 2021 01:00 to 1 October 2026 00:00**, including warm-up. The packaged **22-slot screening universe** includes SAND; the nine frozen strategies use the **21-slot union of the full universe except SAND**. Lifecycle and missingness masks remain explicit. Data documentation | Full coin inventory | File hashes and coverage.

Measurement. Full-period return, elapsed CAGR, drawdown, fees, exposure and turnover used every requested accounting interval. Sharpe was annualized from complete contiguous UTC daily net returns using 365 days; a constant/all-cash series had undefined Sharpe. BTC alpha/beta used the same daily benchmark convention and did not turn a neutral-dollar portfolio into a beta-neutral one. Drawdowns were observed-endpoint measures; intrahour lows and halt interiors were not observed.

Robustness coverage. The retained work included all seven full-period scenarios; both annual slices; exact nonnegative compounded fee roots at both delays; coin/day/month concentration; traded-coin attribution for the full path, each year and the predeclared May event window; matched clocks/buffers; and local parameter/basket perturbations. Finite fee roots were solved from compounded hourly net returns, not gross-return/turnover division. Gross-loss, flat zero-turnover, infinite-tolerance and unavailable states stayed distinct.

Reliability and limits. Complete work-key/scalar identities and saved-file checksums were audited. Independent public-engine reconstruction covered 225 basket-case samples and 1,575 financial scenarios, not every hourly path. Ranking used past liquidity, but inherited universe/source selection remained. The 169-case space was informed by historical research; correlated basket/parameter searches were not search-adjusted discoveries.

The execution model excluded funding, borrow, margin constraints, order-book depth, slippage, market impact and ordinary drift-maintenance trades. Historical volume did not validate capacity. Performance tables use the audited saved strategy paths. The attribution and joint-factor diagnostics use those same paths and their existing factor prices; they do not regenerate signals, targets, trades or selections.

Development-stage conclusion

The defensible development finding was **a cost-and-stability problem**, with some slower configurations worth testing under a rule frozen before validation. It was not “choose the highest development Sharpe” or “trend is proven superior.”

Report 2 follows the development-only selection rule, validation failure diagnosis, reference comparisons and Track B shortlist. Report 3 covers the partial-year 2026 evaluation and subsequent exploratory screening diagnostics. These three reports are the complete reading set.

Companion evidence: compact tables alongside this report retain the displayed denominators and units; the supporting results retain unsuccessful configurations, complete metric tables, accounting diagnostics and provenance. Factor attribution describes the frozen strategies; it is not a separate strategy-selection test.

Per-coin attribution: what each coin contributed, and what joint basket exposure explains

1 January 2022-31 December 2023, all nine unchanged frozen strategies. These explanatory diagnostics separate accounting contribution from statistical sensitivity. Strategy returns, Sharpe, turnover, drawdowns, baskets and decisions remain fixed. The four Track A identities were selected using development; the five Track B identities were later selected using development plus validation. Track B development estimates describe validation-selected identities, not development-only selections.

The overview shows **annual alpha in percent per year**, estimate [95% CI]. Alpha is 365 times the daily arithmetic intercept, **not CAGR**. The BTC-only comparison uses its full-calendar HAC7 estimate; matched BTC and the joint model use the identical complete-case dates with calendar HAC27.

2022-2023 development: all nine frozen strategies

ID	Joint alpha / yr [95% CI]	BTC alpha / yr [95% CI]	Matched BTC alpha / yr [95% CI]	Fit / calendar days	R^2	Max VIF
A01	56.39 [10.70, 102.07]	53.14 [10.15, 96.13]	69.96 [14.63, 125.29]	610/730	0.093	5.46
A02	54.87 [0.78, 108.97]	68.86 [7.81, 129.92]	75.02 [2.57, 147.48]	610/730	0.184	5.46
A03	36.37 [8.21, 64.54]	27.32 [-5.19, 59.82]	27.32 [-7.40, 62.03]	730/730	0.255	5.03
A04	74.89 [-22.18, 171.95]	93.72 [1.51, 185.92]	94.71 [-22.67, 212.10]	610/730	0.118	4.94
B01	-2.00 [-38.41, 34.40]	4.24 [-33.13, 41.60]	4.24 [-34.72, 43.20]	730/730	0.146	5.26
B03	-6.60 [-39.18, 25.98]	4.07 [-30.28, 38.41]	4.07 [-29.71, 37.84]	730/730	0.176	5.99
B04	5.33 [-30.49, 41.16]	27.48 [-34.13, 89.09]	-5.65 [-45.76, 34.46]	610/730	0.135	4.96
B05	7.51 [-20.83, 35.85]	7.34 [-25.19, 39.88]	7.34 [-24.11, 38.80]	730/730	0.544	5.38
B06	19.82 [-36.32, 75.97]	30.88 [-28.41, 90.17]	30.88 [-25.09, 86.85]	730/730	0.174	5.28

How to read this. The joint regression includes an intercept and all eight basket coins simultaneously. A conditional beta is sensitivity holding the other seven factors fixed; it is not a portfolio weight, a causal contribution or proof of market neutrality. R-squared is an in-sample descriptive fit. Comparing joint with matched BTC separates added factors from sample exclusions; the full-calendar BTC-only comparison also uses a different covariance convention.

Uncertainty and missingness. Joint and matched BTC intervals are approximate marginal 95% normal-Wald intervals using Bartlett calendar HAC27 and n/(n-k) correction; HAC7 sensitivity is exported. Missing factor days are excluded from regressions only; gaps are not filled or compressed. The complete strategy and attribution calendars remain intact. Standardized condition >30 or any VIF >10 is a warning, not a reason to drop a coin. Some fits are sensitive to extreme days; high leverage/Cook's distance can coexist with modest collinearity. No event is removed and no model reselected.

Per-coin accounting. Gross contributions and positive allocated fees are both linked with prior **net** strategy NAV; net = gross - fees. Units are whole-period percentage points (pp) of initial capital, not standalone coin returns. The next pages retain all eight slots and their marginal net-contribution intervals.

A01 | basket 4645 | 2022-2023 development

Buffered momentum; 28.0d lookback, 168h rebalance. **Net period return: 165.36%.** The eight net contributions below sum to that unchanged return.

Joint fit **610/730 days**; R-squared **0.093**; standardized condition **6.50**; max VIF **5.46**. Neither declared collinearity warning threshold was crossed. Max leverage **0.279**, max Cook's distance **1.375**: low collinearity alone does not establish stability. Excluded: **2022-05-13 to 2022-09-09 (120 days)**.

BTC-only beta: full-calendar HAC7 **-0.019 [-0.084, 0.047]**; matched-calendar HAC27 **-0.011 [-0.097, 0.074]**. All brackets are marginal 95% intervals.

A01: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-0.36	2.26	-2.62	[-43.61, 30.17]	0.075 [-0.071, 0.221]
ETH	2.91	2.01	0.89	[-37.98, 48.43]	-0.060 [-0.209, 0.090]
SOL	79.56	2.75	76.81	[-12.25, 244.08]	-0.008 [-0.092, 0.076]
LUNA/LUNC	51.76	2.36	49.39	[-22.68, 211.88]	-0.042 [-0.101, 0.018]
FTM/S	58.31	2.24	56.07	[-9.20, 217.33]	0.006 [-0.097, 0.109]
TRX	1.66	2.52	-0.86	[-32.41, 53.91]	0.062 [0.002, 0.122]
LINK	10.77	2.40	8.37	[-58.82, 100.47]	0.065 [-0.002, 0.133]
LTC	-19.86	2.83	-22.69	[-89.62, 20.59]	-0.078 [-0.158, 0.002]

A02 | basket 4645 | 2022-2023 development

Momentum; 28.0d lookback, 72h rebalance. **Net period return: 252.94%.** The eight net contributions below sum to that unchanged return.

Joint fit **610/730 days**; R-squared **0.184**; standardized condition **6.50**; max VIF **5.46**. Neither declared collinearity warning threshold was crossed. Max leverage **0.279**, max Cook's distance **4.224**: low collinearity alone does not establish stability. Excluded: **2022-05-13 to 2022-09-09 (120 days)**.

BTC-only beta: full-calendar HAC7 **-0.027 [-0.088, 0.034]**; matched-calendar HAC27 **-0.015 [-0.105, 0.074]**. All brackets are marginal 95% intervals.

A02: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	10.35	4.40	5.95	[-42.91, 63.45]	0.158 [-0.039, 0.355]
ETH	29.65	4.90	24.74	[-29.78, 114.07]	-0.065 [-0.226, 0.095]
SOL	109.78	5.47	104.31	[-6.75, 373.93]	-0.020 [-0.099, 0.060]
LUNA/LUNC	136.45	4.26	132.19	[-6.28, 930.22]	-0.079 [-0.176, 0.018]
FTM/S	88.81	4.83	83.98	[-3.36, 360.60]	0.035 [-0.061, 0.131]
TRX	-3.43	5.63	-9.06	[-46.61, 87.88]	0.021 [-0.031, 0.073]
LINK	-36.43	5.73	-42.17	[-169.74, 64.25]	0.055 [-0.006, 0.117]
LTC	-41.30	5.71	-47.01	[-187.68, 15.56]	-0.101 [-0.194, -0.007]

A03 | basket 1159 | 2022-2023 development

Reversal; 60.0d lookback, 168h rebalance. **Net period return: 64.38%.** The eight net contributions below sum to that unchanged return.

Joint fit **730/730 days**; R-squared **0.255**; standardized condition **7.07**; max VIF **5.03**. Neither declared collinearity warning threshold was crossed. Max leverage **0.205**, max Cook's distance **0.730**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **0.017 [-0.035, 0.069]**; matched-calendar HAC27 **0.017 [-0.044, 0.079]**. All brackets are marginal 95% intervals.

A03: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-17.32	1.68	-19.00	[-50.02, 22.59]	-0.197 [-0.283, -0.111]
ETH	33.16	1.87	31.29	[-6.75, 85.38]	0.022 [-0.085, 0.129]
SHIB	16.69	2.19	14.50	[-24.33, 58.96]	0.027 [-0.057, 0.112]
DOT	8.30	2.09	6.21	[-26.59, 37.14]	0.009 [-0.112, 0.130]
ADA	-12.02	1.93	-13.95	[-51.09, 29.98]	-0.045 [-0.123, 0.032]
MANA	-21.83	1.76	-23.59	[-85.28, 61.26]	0.102 [0.013, 0.190]
FIL	50.37	2.33	48.04	[-24.72, 151.15]	0.082 [0.029, 0.136]
VET	23.13	2.26	20.88	[-17.10, 75.97]	-0.020 [-0.083, 0.043]

A04 | basket 4444 | 2022-2023 development

Trend long/short; 28.0d lookback, 24h rebalance. **Net period return: 314.22%.** The eight net contributions below sum to that unchanged return.

Joint fit **610/730 days**; R-squared **0.118**; standardized condition **6.61**; max VIF **4.94**. Neither declared collinearity warning threshold was crossed. Max leverage **0.267**, max Cook's distance **1.684**: low collinearity alone does not establish stability. Excluded: **2022-05-13 to 2022-09-09 (120 days)**.

BTC-only beta: full-calendar HAC7 **-0.276 [-0.498, -0.054]**; matched-calendar HAC27 **-0.201 [-0.549, 0.146]**. All brackets are marginal 95% intervals.

A04: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	24.57	4.80	19.76	[-22.98, 160.08]	0.004 [-0.231, 0.239]
ETH	17.82	5.66	12.16	[-44.84, 202.48]	-0.416 [-0.615, -0.217]
SOL	91.50	4.26	87.23	[8.88, 585.91]	0.118 [0.019, 0.217]
LUNA/LUNC	82.66	3.70	78.96	[-20.35, 1138.52]	-0.072 [-0.187, 0.043]
AVAX	52.07	3.62	48.45	[-41.37, 395.91]	-0.062 [-0.192, 0.067]
FTM/S	19.57	4.36	15.21	[-134.20, 299.16]	0.016 [-0.103, 0.136]
FIL	42.72	4.77	37.94	[-19.13, 373.94]	0.107 [0.021, 0.192]
VET	19.20	4.70	14.50	[-40.19, 218.07]	0.047 [-0.109, 0.203]

B01 | basket 2323 | 2022-2023 development

Buffered momentum; 28.0d lookback, 24h rebalance. **Net period return: 0.34%.** The eight net contributions below sum to that unchanged return.

Joint fit **730/730 days**; R-squared **0.146**; standardized condition **6.74**; max VIF **5.26**. Neither declared collinearity warning threshold was crossed. Max leverage **0.493**, max Cook's distance **4.421**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.039 [-0.093, 0.016]**; matched-calendar HAC27 **-0.039 [-0.109, 0.032]**. All brackets are marginal 95% intervals.

B01: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	14.96	3.99	10.97	[-11.08, 33.69]	0.073 [-0.038, 0.184]
ETH	21.09	4.30	16.79	[-10.28, 49.67]	0.014 [-0.107, 0.135]
BNB	-8.14	4.30	-12.44	[-38.26, 7.20]	-0.030 [-0.117, 0.057]
XRP	0.58	3.97	-3.40	[-43.90, 39.20]	-0.066 [-0.155, 0.024]
DOGE	-41.51	5.10	-46.61	[-93.26, -7.60]	0.084 [-0.009, 0.178]
FTM/S	55.69	4.25	51.44	[1.04, 142.73]	-0.024 [-0.123, 0.076]
AXS	3.94	4.37	-0.43	[-39.64, 55.84]	-0.091 [-0.151, -0.031]
VET	-11.81	4.19	-15.99	[-44.33, 15.14]	0.021 [-0.052, 0.095]

B03 | basket 2457 | 2022-2023 development

Momentum; 90.0d lookback, 72h rebalance. **Net period return: 1.05%.** The eight net contributions below sum to that unchanged return.

Joint fit **730/730 days**; R-squared **0.176**; standardized condition **7.12**; max VIF **5.99**. Neither declared collinearity warning threshold was crossed. Max leverage **0.490**, max Cook's distance **0.493**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.018 [-0.077, 0.041]**; matched-calendar HAC27 **-0.018 [-0.098, 0.062]**. All brackets are marginal 95% intervals.

B03: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	9.30	1.10	8.20	[-15.10, 31.46]	0.115 [-0.007, 0.238]
ETH	-24.60	1.36	-25.96	[-58.38, -8.66]	0.029 [-0.064, 0.122]
BNB	-8.87	1.31	-10.17	[-40.15, 12.37]	-0.056 [-0.158, 0.046]
XRP	-21.81	1.48	-23.29	[-56.37, 0.16]	-0.010 [-0.057, 0.038]
FTM/S	41.29	1.18	40.11	[-14.08, 142.99]	0.021 [-0.086, 0.127]
MATIC/POL	-0.73	1.41	-2.15	[-30.79, 34.80]	0.025 [-0.041, 0.091]
AXS	11.18	1.30	9.87	[-34.03, 72.65]	-0.148 [-0.205, -0.091]
LTC	5.56	1.12	4.44	[-13.60, 28.15]	0.017 [-0.043, 0.078]

B04 | basket 6064 | 2022-2023 development

Reversal; 90.0d lookback, 168h rebalance. **Net period return: 44.11%.** The eight net contributions below sum to that unchanged return.

Joint fit **610/730 days**; R-squared **0.135**; standardized condition **6.75**; max VIF **4.96**. Neither declared collinearity warning threshold was crossed. Max leverage **0.255**, max Cook's distance **7.566**: low collinearity alone does not establish stability. Excluded: **2022-05-13 to 2022-09-09 (120 days)**.

BTC-only beta: full-calendar HAC7 **0.009 [-0.063, 0.082]**; matched-calendar HAC27 **-0.008 [-0.093, 0.078]**. All brackets are marginal 95% intervals.

B04: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	-19.97	1.26	-21.24	[-63.80, 19.36]	-0.263 [-0.405, -0.121]
ETH	24.39	1.35	23.05	[-7.40, 76.04]	-0.097 [-0.267, 0.073]
DOT	11.35	1.38	9.97	[-35.21, 35.12]	0.064 [-0.054, 0.182]
LUNA/LUNC	18.40	0.87	17.53	[-46.77, 258.52]	-0.045 [-0.120, 0.030]
AVAX	-14.40	1.51	-15.91	[-72.28, 47.07]	0.031 [-0.121, 0.182]
MANA	-22.30	1.65	-23.95	[-82.22, 48.94]	0.093 [-0.038, 0.224]
FIL	44.40	1.86	42.54	[-41.41, 129.41]	0.095 [0.026, 0.165]
VET	13.96	1.84	12.12	[-24.78, 66.76]	0.048 [-0.035, 0.132]

B05 | basket 2330 | 2022-2023 development

Trend long/cash; 28.0d lookback, 168h rebalance. **Net period return: 12.60%.** The eight net contributions below sum to that unchanged return.

Joint fit **730/730 days**; R-squared **0.544**; standardized condition **6.51**; max VIF **5.38**. Neither declared collinearity warning threshold was crossed. Max leverage **0.489**, max Cook's distance **0.291**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **0.382 [0.273, 0.492]**; matched-calendar HAC27 **0.382 [0.243, 0.521]**. All brackets are marginal 95% intervals.

B05: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	7.37	0.45	6.92	[-3.94, 36.56]	0.057 [-0.013, 0.126]
ETH	4.21	0.47	3.73	[-7.62, 28.55]	0.050 [-0.041, 0.141]
BNB	2.39	0.49	1.89	[-8.31, 17.33]	-0.005 [-0.076, 0.065]
XRP	-1.46	0.63	-2.09	[-13.76, 16.32]	0.047 [0.022, 0.073]
DOGE	-4.01	0.58	-4.60	[-19.88, 15.09]	0.066 [0.030, 0.102]
TRX	-0.08	0.46	-0.53	[-10.99, 16.34]	0.047 [-0.027, 0.122]
GALA	8.21	0.38	7.84	[-8.44, 52.03]	0.031 [-0.003, 0.065]
LINK	-0.10	0.46	-0.56	[-16.90, 33.77]	0.080 [0.033, 0.127]

B06 | basket 2495 | 2022-2023 development

Trend long/short; 28.0d lookback, 168h rebalance. **Net period return: 35.86%.** The eight net contributions below sum to that unchanged return.

Joint fit **730/730 days**; R-squared **0.174**; standardized condition **6.59**; max VIF **5.28**. Neither declared collinearity warning threshold was crossed. Max leverage **0.488**, max Cook's distance **0.226**: low collinearity alone does not establish stability. Excluded: **None**.

BTC-only beta: full-calendar HAC7 **-0.287 [-0.475, -0.100]**; matched-calendar HAC27 **-0.287 [-0.551, -0.023]**. All brackets are marginal 95% intervals.

B06: per-coin accounting and conditional betas

Coin	Gross pp	Fees pp	Net pp	Net 95% CI (pp)	Joint beta [95% CI]
BTC	15.42	1.09	14.33	[-3.69, 61.13]	0.014 [-0.119, 0.147]
ETH	9.40	1.19	8.20	[-11.73, 56.05]	-0.013 [-0.204, 0.178]
BNB	3.41	1.29	2.12	[-15.12, 35.33]	-0.151 [-0.288, -0.014]
XRP	-6.06	1.62	-7.68	[-35.46, 27.20]	-0.015 [-0.060, 0.029]
TRX	-7.18	1.19	-8.37	[-44.29, 11.75]	-0.019 [-0.169, 0.130]
MANA	7.64	0.88	6.76	[-27.61, 67.42]	-0.076 [-0.204, 0.052]
GALA	32.13	0.89	31.24	[-1.63, 124.54]	-0.063 [-0.110, -0.016]
LINK	-9.59	1.16	-10.75	[-41.39, 35.18]	0.042 [-0.060, 0.144]

Full precision, sensitivity and multiplicity

Per-coin gross, fee and net attribution with all three confidence intervals | Joint alpha/betas, HAC7 sensitivity, global Holm243 p-values and Bonferroni243 intervals | Joint model diagnostics and individual VIFs | Full-calendar BTC-only estimates and intervals | Matched BTC-only coefficients and intervals | Regression calendar and exclusions.

Attribution uses **4,000 seeded, synchronized circular 28-day calendar-block draws** across all nine strategies and eight coins. Daily gross/fee vectors are resampled together and relinked using each draw's **new prior NAV**; signals, targets and trades are not regenerated and no block-boundary fee is invented. Any invalid draw makes the affected interval unavailable; none is silently dropped. Gross/fee intervals and invalid-draw counts remain in the CSV.

The coefficient correction spans **243 predeclared diagnostic slots: 27 joint alphas + 216 joint betas** across all three periods. Displayed intervals are marginal; approximate Bonferroni familywise intervals and Holm-adjusted p-values are exported. This corrects only that conditional coefficient family, **not the original universe, configuration, period or strategy-selection search.** Attribution intervals are marginal and not multiplicity-adjusted. No significance row changes the frozen shortlist or crowns a new winner. LUNA/LUNC is legacy Terra, not Terra 2.0; FTM/S and MATIC/POL retain the original economic slots.